

## 03.08 Applications of Linear Programming

Common problems and solution methods

### Rule guide: recognize the task first

| Recognize                                                    | Apply these rules                                                                                              |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Linear Relationships And Constraints                         | Choose the representation that matches the wording: equation, inequality, system, or objective function.       |
| The expression contains the chapter's key notation or model. | Write the known quantities, choose the matching formula, calculate carefully, and check the result in context. |

### Example 1: Identify the model

#### Problem

$$2x + y = 7$$

#### How to solve it

Write the known quantities and restrictions.

$$y = mx + b$$

Check the result against the original statement.

### Example 2: Apply the key rule

#### Problem

$$y > -2x + 3$$

#### How to solve it

Write the known quantities and restrictions.

$$ax + by = c$$

Check the result against the original statement.

### Example 3: Check the result

#### Problem

$$P = 4x + 3y$$

#### How to solve it

Write the known quantities and restrictions.

$$P = ax + by$$

Check the result against the original statement.